BIKE SALES DATA ANALYSIS – EXCEL PROJECT

Data Cleaning, Analysis, Charts

1. Introduction

This project analyzes a Bike Sales customer dataset using Microsoft Excel. The work covers data cleaning, descriptive analysis, visualization and interactive filter-style controls. The purpose is to understand customer demographics and bike purchase behavior from the available records.

2. Dataset Description

The original workbook contains 1028 rows and 13 columns: ID, Marital Status, Gender, Income, Children, Education, Occupation, Home Owner, Cars, Commute Distance, Region, Age and Purchased Bike. The source contains a blank record and a zero ID record at the end of the data.

3. Data Cleaning

The cleaning step removed rows without a valid customer ID and removed the zero-ID record. Duplicate records were also checked and removed if present. The resulting cleaned dataset contains 1000 valid customer records.

Before Cleaning – Bike Sales Dataset

ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distanc	Region	Age	Purchased Bike
12496.0	M	F	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	42.0	No
24107.0	M	M	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles	Europe	43.0	No
14177.0	M	M	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles	Europe	60.0	No
24381.0	S	M	70000.0	0.0	Bachelors	Professional	Yes	1.0	5-10 Miles	Pacific	41.0	Yes
25597.0	S	M	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles	Europe	36.0	Yes
13507.0	M	F	10000.0	2.0	Partial College	Manual	Yes	0.0	1-2 Miles	Europe	50.0	No
27974.0	S	M	160000.0	2.0	High School	Management	Yes	4.0	0-1 Miles	Pacific	33.0	Yes
19364.0	M	M	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles	Europe	43.0	Yes
22155.0	M	M	20000.0	2.0	Partial High School	Clerical	Yes	2.0	5-10 Miles	Pacific	58.0	No
19280.0	M	M	120000.0	2.0	Partial College	Manual	Yes	1.0	0-1 Miles	Europe	40.0	Yes
22173.0	M	F	30000.0	3.0	High School	Skilled Manual	No	2.0	1-2 Miles	Pacific	54.0	Yes
12697.0	S	F	90000.0	0.0	Bachelors	Professional	No	4.0	10+ Miles	Pacific	36.0	No

Before cleaning

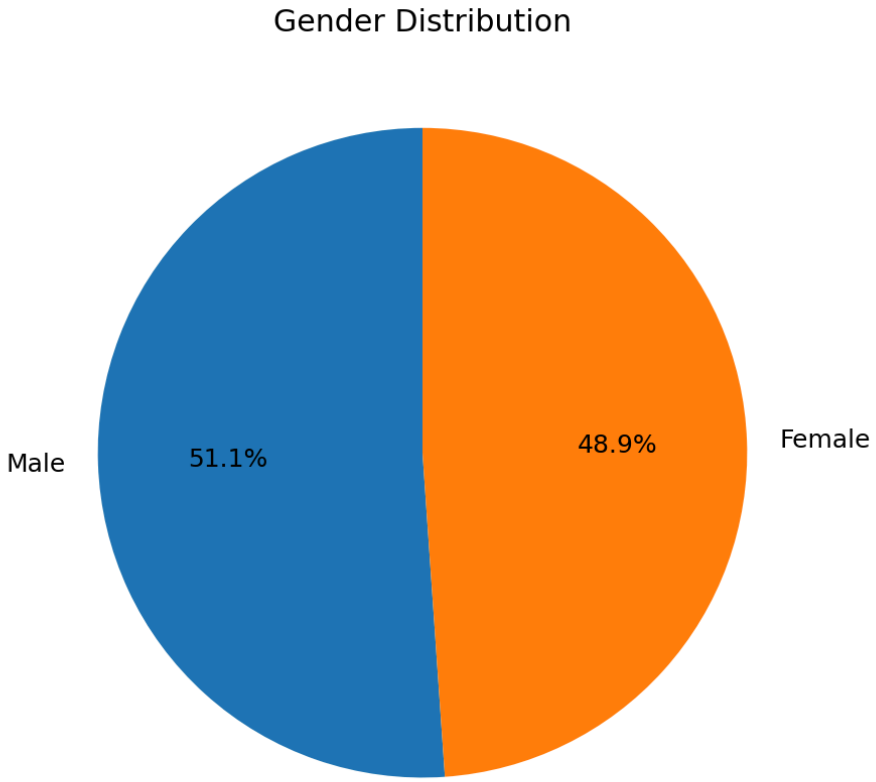
After Cleaning – Bike Sales Dataset

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After cleaning

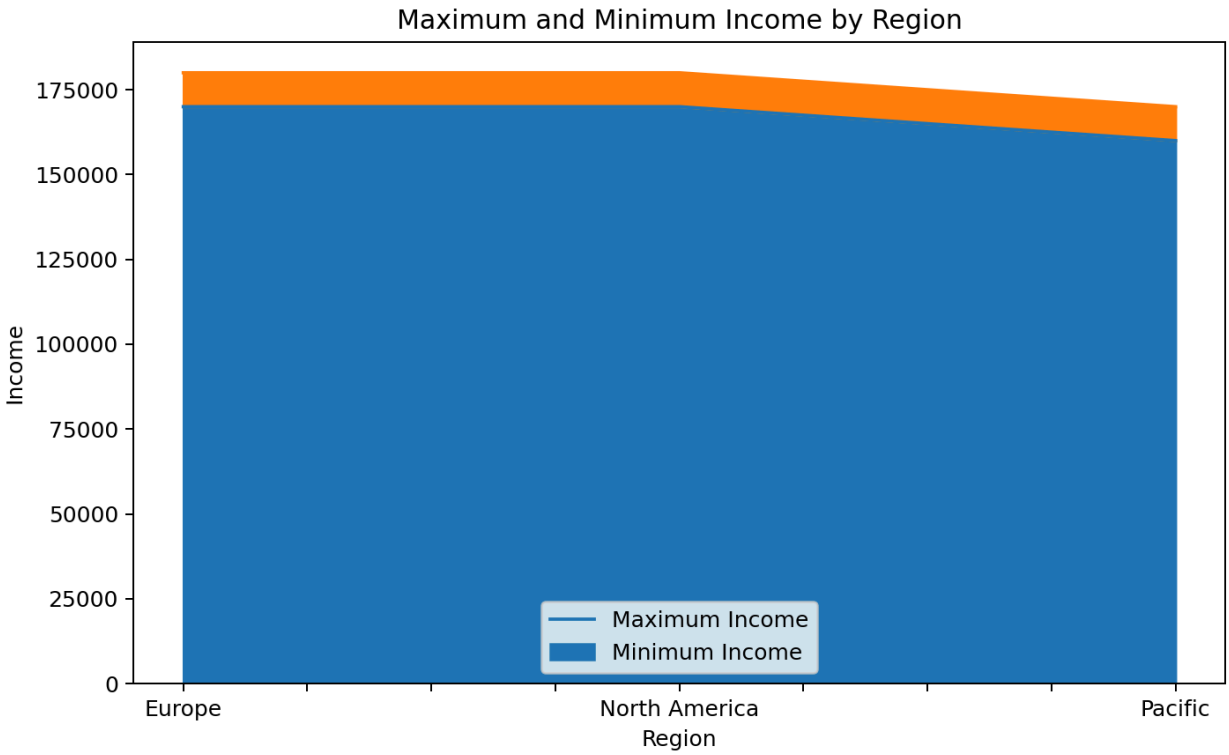
4. Analysis and Visualization

The cleaned data has an average customer age of 44.16 years. There are 511 male and 489 female records. The maximum recorded income is $170,000, while the minimum recorded income is $10,000.



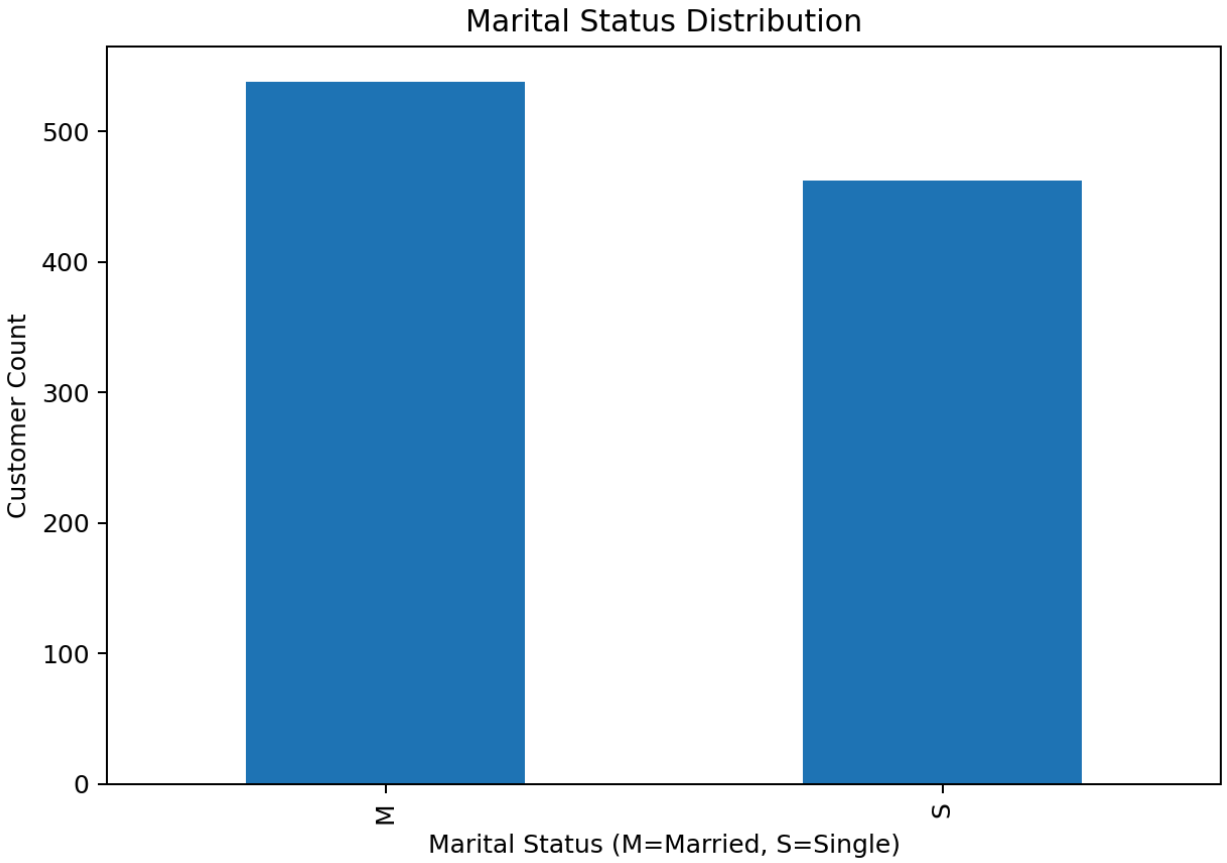
Gender distribution pie chart

The pie chart summarizes the proportion of male and female customers in the cleaned dataset.



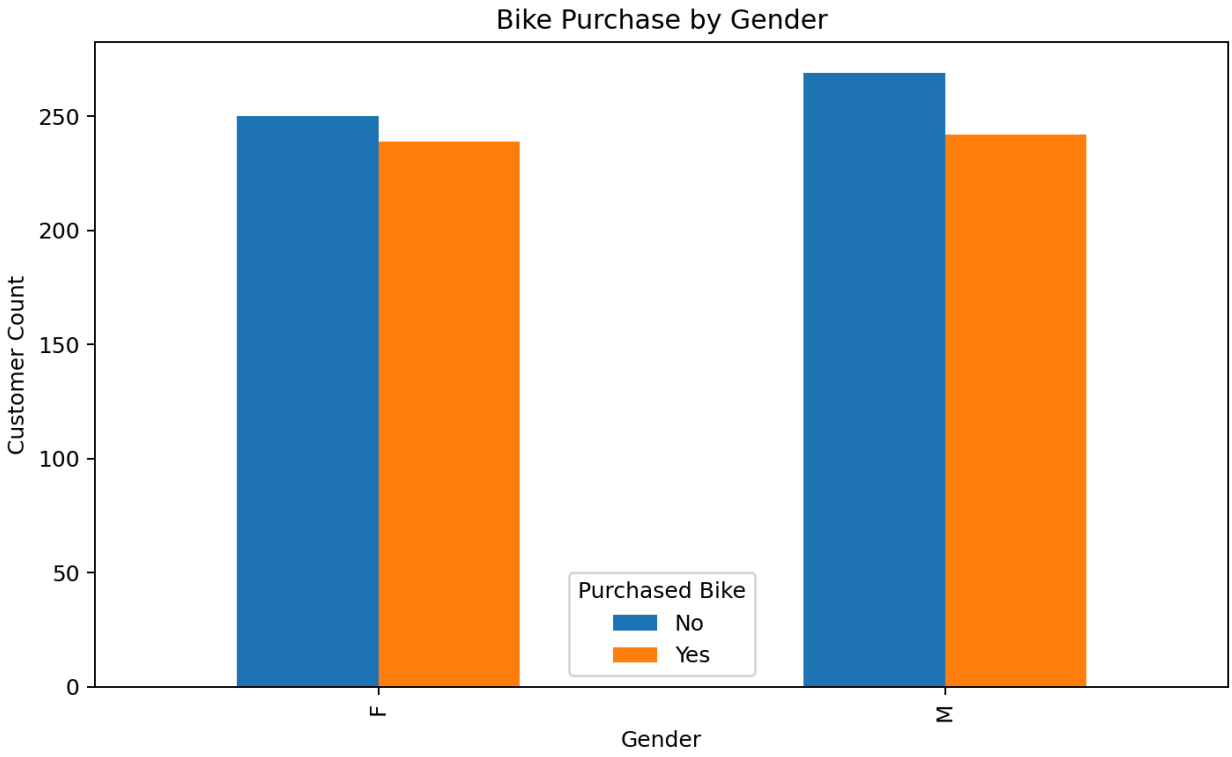
Maximum and minimum income by region – area chart

The area chart compares the maximum and minimum recorded income across regions.

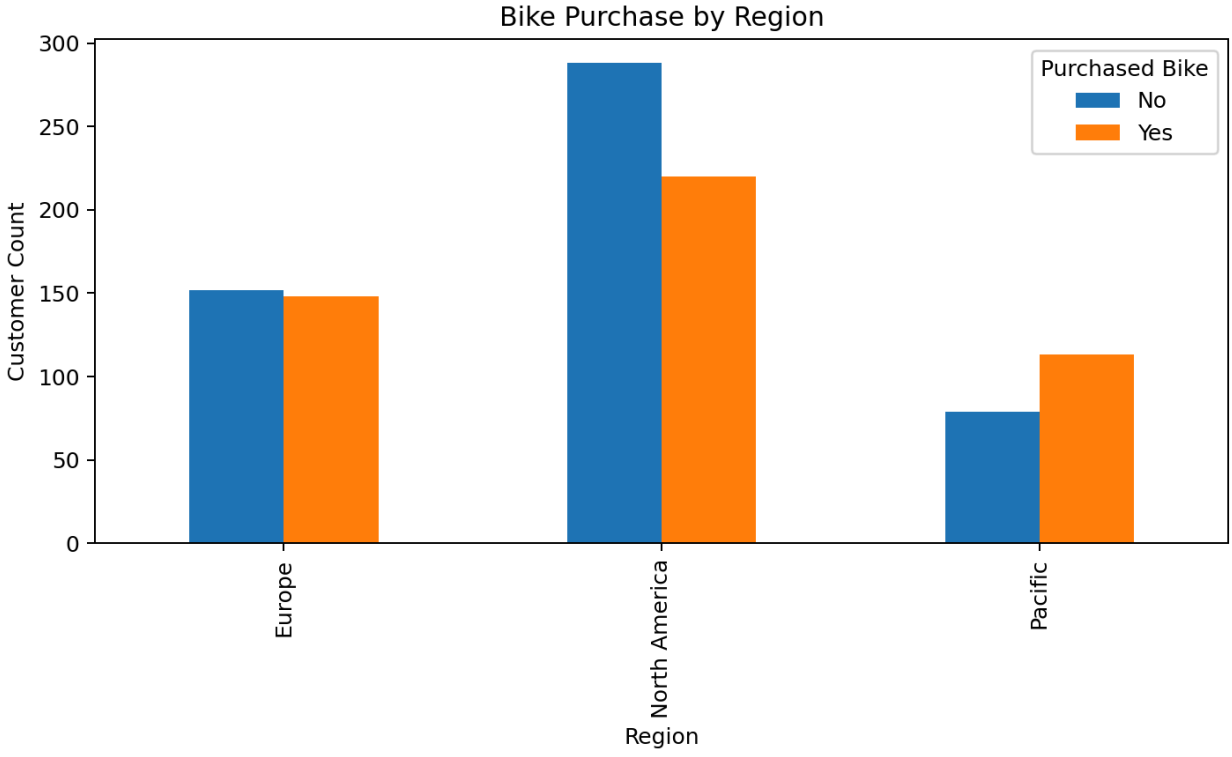


Marital status bar chart

The bar chart shows the number of customers in the two marital-status categories used in the dataset: M (Married) and S (Single).



Bike purchase by gender



Bike purchase by region

6. Key Findings

- Cleaned dataset size: 1000 valid customer records.
- Average customer age: 44.16 years.
- Male records: 511; Female records: 489.
- Income range: $10,000 to $170,000.
- Bike purchase status can be compared across gender and region using the prepared charts.

7. Conclusion

The Bike Sales dataset provides customer demographic, financial and transportation-related information. Cleaning the records and presenting the information through charts makes the dataset easier to interpret. The prepared Excel workbook contains the cleaned data, analysis tables, charts and filter-style dashboard controls for further exploration.

8. Files Included

The accompanying Excel workbook contains Cleaned Data, Analysis and Dashboard sheets. The Dashboard sheet contains slicer-style dropdown controls designed for mobile Excel compatibility.